MIL-D-45649(Ord)
15 June 1961

SUPERSEDING
FA(FC)-PD-529
14 August 1959

MILITARY SPECIFICATION

DEFLECTION UNIT: 8644083

1. SCOPE

1.1 This specification covers one type of deflection unit used in the Gun Direction Computers: M15 and M15C, which are parts of Field Artillery Fire Control Systems: M35 and M35C. The deflection unit is a functional plug-in subassembly performing two primary functions as described in 6.1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of
MIL-I-45208	Inspection Requirements, General Specification for
MIL-I-45645	Amplifier Assembly, Dual Booster: 8644111

STANDARDS

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1220

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DRAWINGS

Ordnance Corps

F8644083	Deflection Unit
F8644090	Electrical Schematic, Deflection Unit

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one deflection unit, representative of production, shall be made available to the Government to determine that the deflection unit meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The deflection unit shall be manufactured in accordance with Drawing F8644083. "Deflection Unit," and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical, characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.3.1 Ground potential.- The design and construction of the deflection unit shall insure that all external parts exclusive of electrical connections, are at ground potential at all times.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electrical
(f) Workmanship

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3.5 Environmental.

3.5.1 Vibration.- The deflection unit shall be subjected to a simple harmonic motion with an amplitude of 0.015-inch (0.03-inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the deflection unit shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.5.2 Operating temperatures.- The deflection unit shall meet the requirements of 3.6 through 3.11 when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.5.3 Storage temperatures.- The deflection unit shall show no evidence of physical failure and shall meet the requirements of 3.6 through 3.11 at room temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 65 degrees Fahrenheit.

3.6 Input knob.- The input knob shall be of the slip-clutch type. When depressed the input shall be capable of being turned to record values on the SUM counter. Upon release of the knob, the knob shall return to the locked position.

3.6.1 Torque.- The torque required to rotate the knob shall be between 10 and 20 ounce-inches. Slippage of the knob shall occur between 2 and 3 pound-inches.

3.7 SUM counter.- The SUM counter shall be capable of rotating from zero to at least 3995 mils and not more than 4000 mils, as determined by rotation of the knob.

3.8 AZIMUTH counter.- The AZIMUTH counter shall indicate from zero to 6399 mils and shall be capable of rotating without limits in either direction from zero.

3.9 DEFLECTION counter.- The DEFLECTION counter shall indicate from zero to 3199 mils and shall be capable of rotating without limits in either direction from zero.

3.10 Backlash.- The backlash in the counter gear train shall be no greater than 1/2 mil of azimuth and 1 mil of deflection.

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3.11 Electrical.

3.11.1 Circuitry.- Circuitry shall be in accordance with electrical schematic, Drawing F8644090.

3.11.2 Deflection unit motor operation.- When properly energized, clockwise rotation of the deflection unit motor shall be indicated as increasing azimuth on the azimuth counter and decreasing deflection on the deflection counter. The indicated change in AZIMUTH and DEFLECTION counters produced by the deflection unit motor shall have no effect on the SUM counter reading.

3.11.3 Accuracy and synchronization of Azimuth Resolver, B1.- With a motor control voltage of 0 to 28 volts, 400 cps; motor reference voltages of 115 volts, 400 cps; and other voltages depicted on the electrical schematic, the Azimuth Resolver, B1 electrical output signals shall be proportional to the triogonometric resolution of the input signals as determined by the angular positions of the resolver shaft when loaded with 250K ohms plus or minus 0.5 percent across each motor winding. The output voltages and counter readings (mils) specified in Table I shall be within the tolerances and readings, excluding backlash:

TABLE I

S3-S1 Input (volts)	Azimuth Counter (mils)	Azimuth Output (volts)	Range Output (volts)	Tolerance (millivolts)
15	0	0	-15	± 25
15	1600	-15	0	± 25
15	3200	0	15	± 25
15	4800	15	0	± 25

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TABLE I (Cont'd)

S2-S4 Input (volts)	Azimuth Counter (mils)	Azimuth Output (volts)	Range Output (volts)	Tolerance (millivolts)
15	0	15	0	± 25
15	1600	0	-15	± 25
15	3200	-15	0	± 25
15	4800	0	15	± 25

NOTE 1: Rotating the input knob shall have no effect on the AZIMUTH counter, nor resolver shafts if azimuth servo is energized and held.

NOTE 2: Minus signs indicate output signal 180 degrees out of phase to input signal which is applied through a Dual Booster Amplifier Assembly (Specification MIL-A-45645). Resolver body and amplifier gain controls may be varied slightly for smallest average error.

3.11.4 Accuracy and synchronization of Deflection Wind Resolver, B2.- With a motor control voltage of 0 to 28 volts, 400 cps; motor reference voltages of 115 volts 400 cps; and other voltages as depicted on the electrical schematic, the Deflection Wind Resolver, B2 electrical output signals shall be proportional to the trigonometric resolution of the input signals as determined by the angular positions of the resolver shaft when loaded with 250K ohms plus or minus 0.5 percent across each rotor winding. The output voltages and counter readings (mils) specified in Table II shall be within the tolerances specified as a result of introducing the respective voltage and readings, excluding backlash:

TABLE II

S3-S1 Input (volts)	Azimuth Counter (mils)	Deflection Wind Output (volts)	Range Wind Output (volts)	Tolerance (millivolts)
15	0	0	15	± 25
15	1600	-15	0	± 25
15	3200	0	-15	± 25
15	4800	15	0	± 25

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TABLE II (Cont'd)

S2-S4 Input (volts)	Azimuth Counter (mils)	Deflection Wind Output (volts)	Range Wind Output (volts)	Tolerance (millivolts)
15	0	15	0	± 25
15	1600	0	15	± 25
15	3200	-15	0	± 25
15	4800	0	-15	± 25

NOTE: Notes 1 and 2 of Table I shall apply.

3.12 Alinement and calibration.- Alinement and calibration of all deflection units shall be determined using the same power source to insure uniformity of product. Simulated voltages of △E, (east wind) and △N (north wind) shall be supplied through suitable resolver booster amplifiers from a power source maintained as a standard.

3.13 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General provisions.- The quality assurance provisions of referenced documents, to the extent specified herein, together with provisions of this section are applicable to inspection performance by the contractor and for Government verification.

4.1.1 The contractor shall provide an inspection system in accordance with Specification MIL-I-45208.

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4.2 Production model.- Unless otherwise specified in the contract, one deflection unit shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.2.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection, and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.3 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.4 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance to drawing requirements prior to acceptance. These characterisitcs are defined as drawing requirements affecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a precision requirement that have a bearing on the functional accuracy and performance of the deflection unit. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.4.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table III.

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TABLE III - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.4.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.5 Material.- Control for requirements of material shall be in accordance with Specification MIL-I-45208.

4.6 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.6.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6.2 Accuracy requirements for special testing devices.

4.6.2.1 Device to test vibration requirements.- The device shall consist of a vibrating machine which will produce motion as specified. It shall be equipped with a medium capable of holding the unit securely to the vibrating machine in the specified positions.

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4.6.2.2 Device to test torque requirements.- The device shall consist of a torque wrench type tester which will permit reading in both the clockwise and counterclockwise directions with a minimum range of from zero to twenty-five ounce-inches, the accuracy of which shall be within plus or minus one-quarter ounce-inch throughout the entire zone of excursion. The device shall be equipped with an adapter to accommodate the knob. The device for testing slippage of the knob shall be similar, but with a minimum range of from zero to five pound-inches, the accuracy of which shall be within plus or minus one-quarter pound-inch throughout the entire zone of excursion. The device shall accommodate the same adapter used for the above test for the knob.

4.7 Tests.

4.7.1 Classification of defects.- The classification of defects in Tables IV and V shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.7.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE IV- CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 4.0%	Req't Par.	QA. Par.	Test Method
101. Post vibration	3.5.1	4.7.2	Device, Special Testing (DST)
102. Operating temperatures	3.5.2	4.7.3	DST

MINOR: NONE DEFINED

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TABLE V - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%		Req't Par.	QA. Par.	Test Method
101.	Input knob	3.6	4.7.5	Visual
102.	Torque	3.6.1	4.7.6	Device, Special Testing (DST)
103.	Sum counter	3.7	4.7.7	Visual
104.	Azimuth counter	3.8	4.7.8	Visual
105.	Deflection counter	3.9	4.7.9	Visual
106.	Backlash	3.10	4.7.10	Standard Measuring Equipment (SME)
107.	Circuitry	3.11.1	4.7.11	Visual
108.	Deflection unit motor operation	3.11.2	4.7.12	Visual
109.	Azimuth resolver, B1	3.11.3	4.7.13	SME
110.	Deflection wind resolver, B2	3.11.4	4.7.14	SME
111.	Identification marking	3.13	4.7.15	Visual

4.7.2 Vibration.- This test shall be conducted in accordance with Method 201A for Standard MIL-STD-202, except that the time period for applied motion in each direction shall be as specified herein. Tests and performance measurements prior to or during vibration are not required for final acceptance. Subsequent to vibration, the deflection unit shall be subjected to a visual-tactile examination for damage. The unit shall then be subjected to all the tests in this specification.

4.7.3 Operating temperatures.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond condition "C" or raised beyond condition "G". When the deflection unit is stabilized at each indicated temperature, it shall be subjected to the tests specified in 4.7.5 to 4.7.15 inclusive, and shall meet the requirements specified.

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4.7.4 Storage temperature.- This test shall be conducted in accordance with Procedure II, condition "B" and "H" of Specification MIL-F-13926. Upon completion of the storage temperature cycling, the deflection unit shall be subjected to tests specified in 4.7.5 to 4.7.15 inclusive, and shall meet the requirements specified. This test shall be performed on the production model only.

4.7.5 Input knob.- The input knob shall be inspected for its functional application. The knob shall be unlocked when depressed, and locked when released.

4.7.6 Torque.- This test shall be performed with the aid of special testing devices, affixed to the input knob, that shall be capable of applying torque to the knob within the ranges specified. The torque required for rotation of the input knob and for clutch slippage shall be within the values specified.

4.7.7 SUM counter.- The SUM counter shall be rotated through the mil values specified and must read within the required mils.

4.7.8 AZIMUTH counter.- The AZIMUTH counter shall be rotated by means of the input knob through the required mil values. A visual check will ascertain if the requirements are met.

4.7.9 DEFLECTION counter.- The DEFLECTION counter shall be rotated by means of the input knob through the required mil values. A visual check will ascertain if the requirements are met.

4.7.10 Backlash.- Energize the unit and advance the counter to some reading and record the output signal for that reading. Continue rotation of the counter in the same direction for one revolution, then reverse rotation and return until the output signal repeats the recorded reading. The counter shall have repeated its reading within the specified tolerance on each counter.

4.7.11 Circuitry.- An electrical point-to-point check shall be made using standard measuring equipment to assure that the values of the circuits conform to the electrical schematic.

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4.7.12 Deflection unit motor operation.- With the energized deflection unit motor rotating in a clockwise direction, a visual check shall be made to be assured that the three (3) counters react as specified.

4.7.13 Accuracy and synchronization of Azimuth Resolver, B1.- Energize the unit as specified and then check the output voltages and counter readings listed in Table I. These readings must be within the tolerances specified.

4.7.14 Accuracy and synchronization of Deflection Wind Resolver, B2.- Energize the unit as specified and then check the output voltages and counter readings specified in Table II. These readings must be within the tolerances specified.

4.7.15 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The deflection unit shall be cleaned, preserved, packed, and marked in accordance with Specification MIL-F-14619 P8519. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The deflection unit is a functional plug-in subassembly of the Gun Direction Computers: M15 and M15C intended to perform two functions:

(a) To indicate target azimuth and gun deflection on counters
(b) To resolve wind information into components of range and deflection winds

6.1.1 The azimuth resolver receives electrical signals of △ E (easting direction from gun battery to target) and △ N (northing direction from gun battery to target) through resolver booster amplifiers and converts them into output signals of azimuth and range. The signal for the azimuth winding will be transmitted through the computer to a nulling servo amplifier which will then transmit control voltages back to a motor in the deflection unit. This motor drives a gear train that positions the azimuth and wind deflection resolver shafts and a azimuth counter to correspond to target azimuth referenced to north.

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6.1.2 The wind deflection resolver receives electrical signals of east wind and north wind through resolver booster amplifiers and converts these into output signals of range wind and deflection wind. In addition, the resolvers generate from compensation windings, control voltages to be used as feedback voltages for the resolver booster amplifiers. The electrical signals of both resolvers are transmitted within the Gun Direction Computers: M15 and M15C.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title number and date of this specification
(b) Selection of applicable level of preservation packaging and packing
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Standard measuring equipment (SME).- Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.4.2 Special testing device (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.4.3 Tactile.- Skillful manipulation or sensing.

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6.4.4 Production model.- A model in its final mechanical and electrical form of final production design made by production methods, tools, jigs and fixtures.

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